VLSI Devices
Lecture 5

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L5

General Q_s-ϕ_s relation (1)

Inversion charge

- A generation relation for $Q_s = Q_d + Q_i$
  - The Poisson equation

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_{si}}[p(x) - n(x) - N_a]$$

Taur, Eq. (4.9)

General Q_s-ϕ_s relation (2)

- Following Prof. Taur's notation,

For a while, $\phi(\infty) = -\phi_B$ is used as the reference value. Therefore,

$$n(x) = n_i \exp\left(\frac{q\phi(x)}{k_B T}\right) \rightarrow n(x) = n(\infty) \exp\left(\frac{q\phi(x)}{k_B T}\right)$$ Taur, Eq. (4.12)

$$p(x) = n_i \exp\left(-\frac{q\phi(x)}{k_B T}\right) \rightarrow p(x) = p(\infty) \exp\left(-\frac{q\phi(x)}{k_B T}\right)$$ Taur, Eq. (4.11)

– The Poisson equation

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_{si}}\left[N_a\left(\exp\left(-\frac{q\phi}{k_B T}\right) - 1\right) - \frac{n_i^2}{N_a}\left(\exp\left(\frac{q\phi}{k_B T}\right) - 1\right)\right]$$

Taur, Eq. (4.13)

General Q_s-ϕ_s relation (3)

- Multiplying $\frac{d\phi}{dx}dx$,
  - The Poisson equation

$$\frac{d\phi}{dx}d\left(\frac{d\phi}{dx}\right) = -\frac{q}{\epsilon_{si}}\left[N_a\left(\exp\left(-\frac{q\phi}{k_BT}\right)-1\right)-\frac{n_i^2}{N_a}\left(\exp\left(\frac{q\phi}{k_BT}\right)-1\right)\right]d\phi$$

  - Integrate the above equation.

$$\int_0^{-E_x(x)}\frac{d\phi}{dx}d\left(\frac{d\phi}{dx}\right) = -\frac{q}{\epsilon_{si}}\int_0^{\phi(x)}\left[N_a\left(\exp\left(-\frac{q\phi}{k_BT}\right)-1\right)-\frac{n_i^2}{N_a}\left(\exp\left(\frac{q\phi}{k_BT}\right)-1\right)\right]d\phi$$

Taur, Eq. (4.14)

General Q_s-ϕ_s relation (4)

- (Square of) Electric field
  - From $\frac{1}{2}E_x^2(x) = -\frac{q}{\epsilon_{si}}\left[-N_a\frac{k_BT}{q}\exp\left(-\frac{q\phi}{k_BT}\right) - N_a\phi + N_a\frac{k_BT}{q} - \frac{n_i^2}{N_a}\frac{k_BT}{q}\exp\left(\frac{q\phi}{k_BT}\right) + \frac{n_i^2}{N_a}\phi + \frac{n_i^2}{N_a}\frac{k_BT}{q}\right]$, we get

$$E_x^2(x) = \frac{2k_BTN_a}{\epsilon_{si}}\left[\left(\exp\left(-\frac{q\phi}{k_BT}\right) + \frac{q\phi}{k_BT} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi}{k_BT}\right) - \frac{q\phi}{k_BT} - 1\right)\right]$$

Taur, Eq. (4.15)

General Q_s-ϕ_s relation (5)

- At $x = 0$, we have $\phi(0) = \phi_s$.
  - Then,

$$E_s^2 = \frac{2k_B T N_a}{\epsilon_{si}}\left[\left(\exp\left(-\frac{q\phi_s}{k_B T}\right) + \frac{q\phi_s}{k_B T} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_B T}\right) - \frac{q\phi_s}{k_B T} - 1\right)\right]$$

General Q_s-ϕ_s relation (6)

- At $x = 0$, we have $\phi(0) = \phi_s$.
  - From $Q_s = -\epsilon_{si} E_s$,

$$Q_s = \pm\sqrt{2\epsilon_{si} k_B T N_a}\left[\left(\exp\left(-\frac{q\phi_s}{k_B T}\right) + \frac{q\phi_s}{k_B T} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_B T}\right) - \frac{q\phi_s}{k_B T} - 1\right)\right]^{1/2}$$

Taur, Eq. (4.16)

Homework#5

- Due: 08:00 on March 23
- Submit a PDF report through the GIST LMS system.
  - 1) First, draw calculate Q_s (or E_s) as a function of ϕ_s.
  - Use the parameters for our example. (Gate workfunction = 4.1 eV. 2-nm-thick oxide. 10^{18} cm^{-3}.)

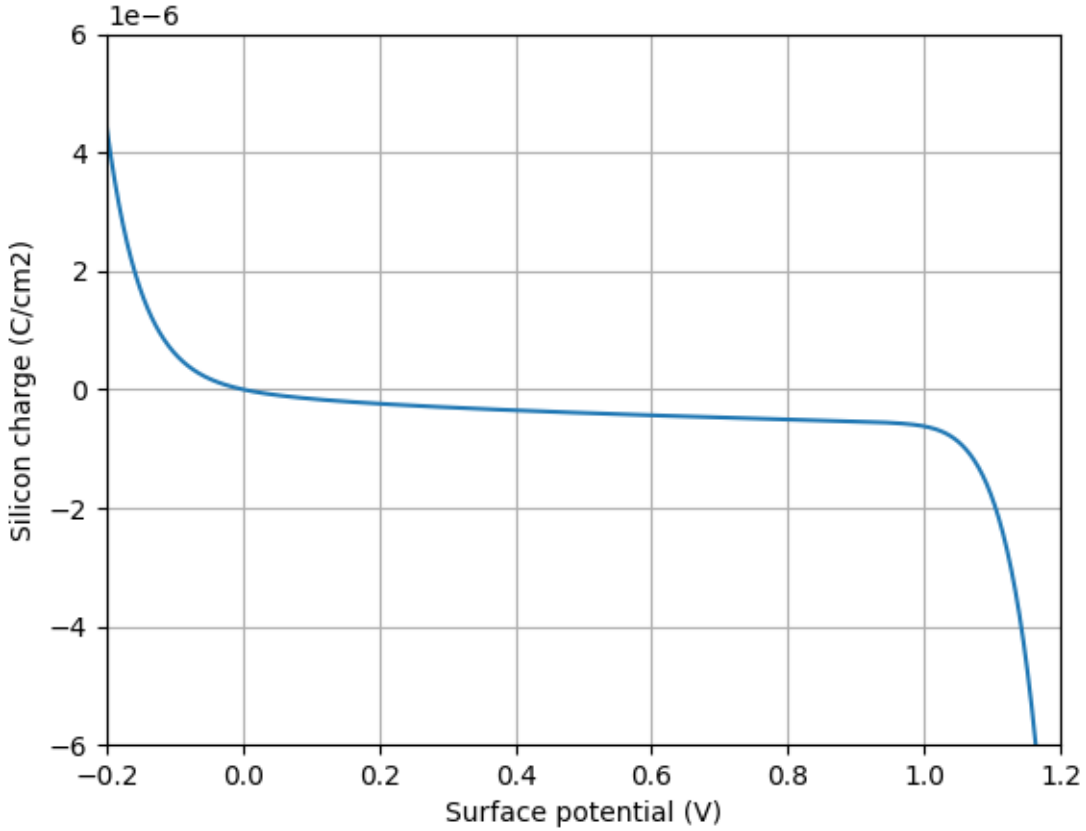
  - 2) From E_s, you can also calculate $V_{ox} = t_{ox}E_{ox} = t_{ox}\frac{\epsilon_{si}}{\epsilon_{ox}}E_s$.
  - Remember that

$$V_g - V_{fb} = \phi_s + V_{ox}$$

  - Draw Q_s as a function of V_g.

Answer of Homework#5 (1)

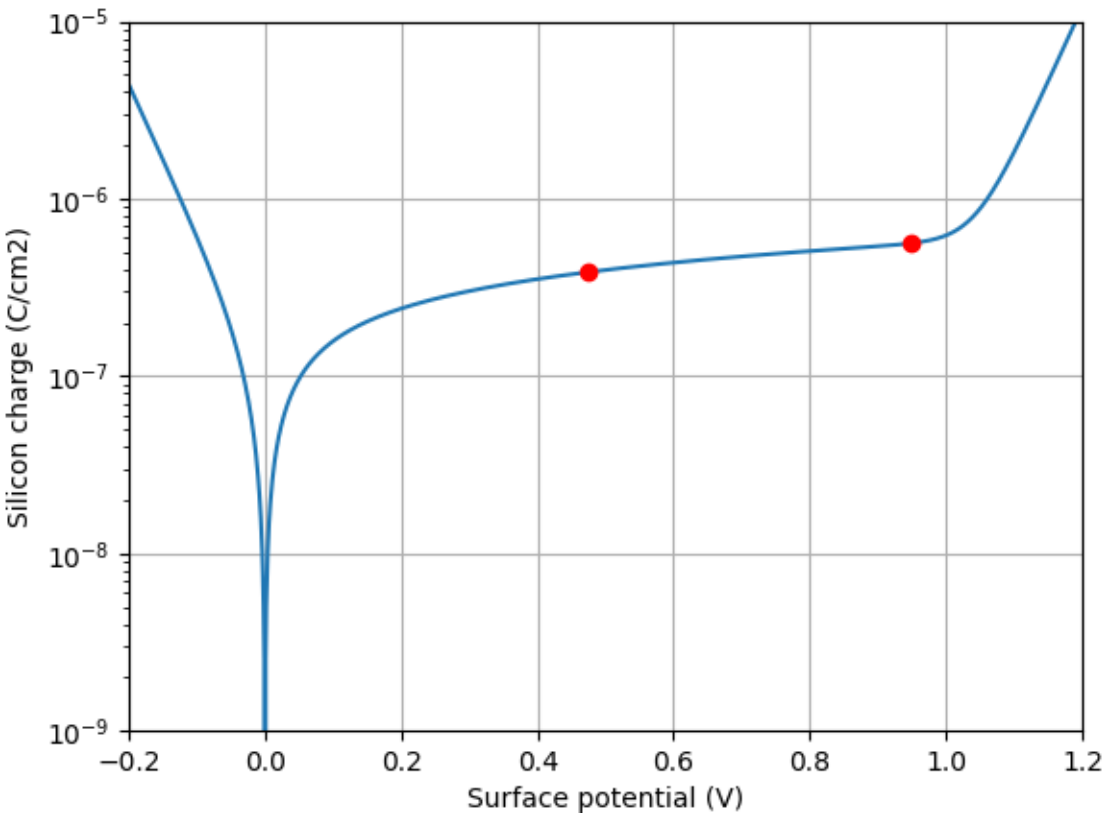
- Draw Eq. (4.16). Use n_i = 1.075X10^{10} cm^{-3}.
  - In this example, $\phi_B = -\phi(\infty)$ is 0.474 V.
  - Linear scale
  - Note) Q_s = 0 @ ϕ_s = 0
  - Note) Q_s < 0 @ ϕ_s > 0
  - Note) Q_s > 0 @ ϕ_s < 0



Answer of Homework#5 (2)

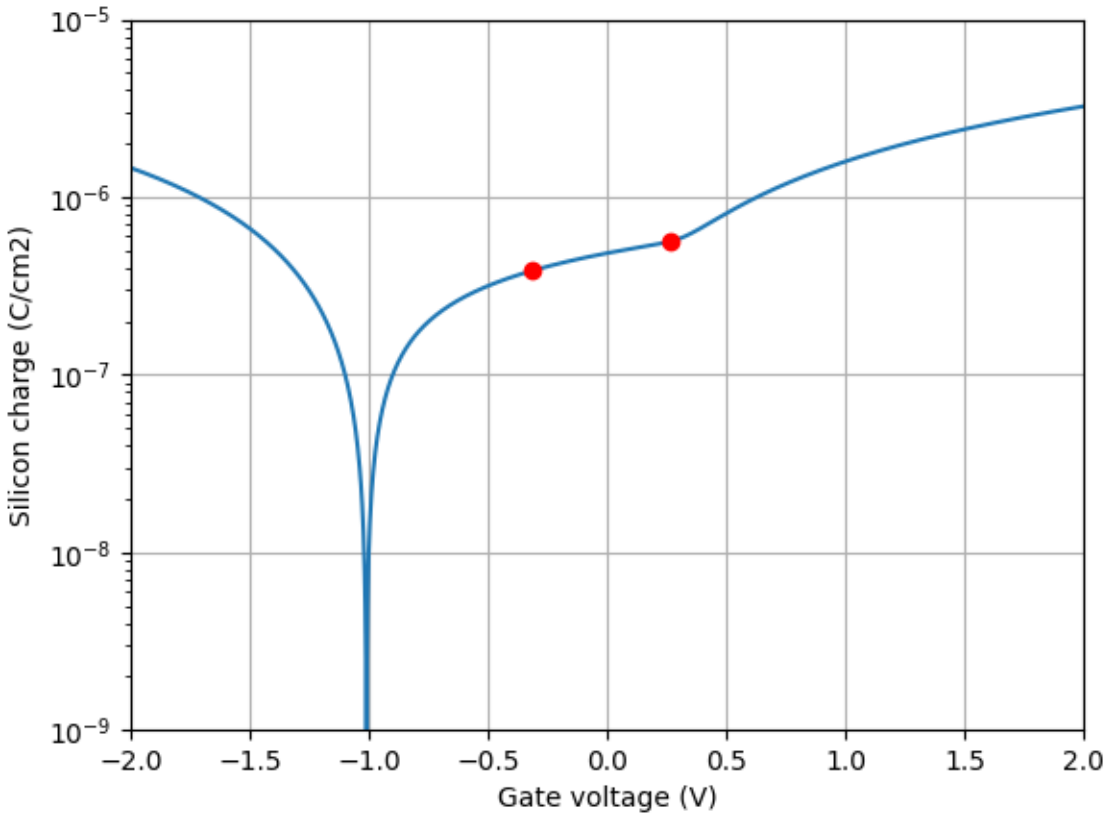
- Semi-logarithmic scale (Absolute value)
  - Accumulation
  - Depletion
  - Weak inversion
  - Strong inversion ($\phi_s = 2\phi_B$)
  - Remember that
  $\phi_s = \phi(0) - \phi(\infty) = \phi(0) + \phi_B$.
  - It means $\phi(0) = \phi_B$.

(in the general convention)



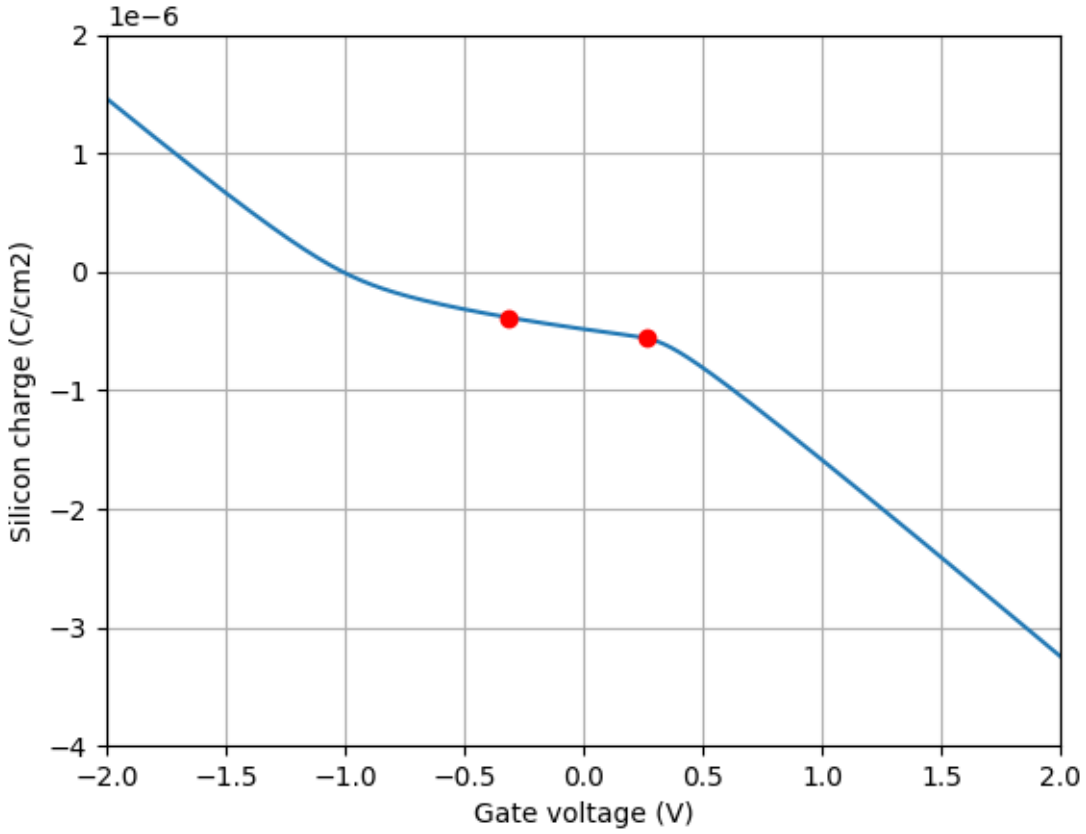
Answer of Homework#5 (3)

- Calculate V_g by using $V_g - V_{fb} = \frac{-Q_s}{C_{ox}} + \phi_s$.
  - Draw Q_s as a function of V_g.



Answer of Homework#5 (4)

- Linear scale
  - In the strong inversion region, Q_s varies linearly.



Thank you!